TRIPATHI STUDY ZONE
JEE & NEET
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CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 10

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three sections in the question paper consisting of Physics (Q.no.1 to 25), Chemistry (Q.no.26 to 50) and Mathematics (Q. no.51 to 75). Each section is divided into two parts, Part I consists of 20 multiple choice questions & Part II consists of 5 Numerical value type Questions.
5. There will be only one correct choice in the given four choices in Part I. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice for Part I Questions and zero mark will be awarded for not attempted question. For Part II Questions 4 marks will be awarded for correct answer and zero for unattempted and incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

PART-I (Multiple Choice Questions)

1. Drops of water fall from the roof of a building 9m high at regular intervals of time. The first drop reaching the ground at the same instant fourth drop starts its fall. What are the distances of second and third drops from roof ?

(a) 6m & 3m (b) 4m & 2m

(c) 6m & 2 m (d) 4m & 1m

2. A 55 gm tennis ball strikes the ground and rebounds back. If its velocities are 5 m/s and 4 m/s before and after the collision respectively then the impulse imparted will be

(a) – 0.495 Ns (b) 0.495 Ns

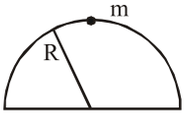
(c) 495 Ns (d) – 495 Ns

3. The time taken by a body in sliding down a rough inclined plane of angle of inclination 45° is n times the time taken by the same body in slipping down a similar frictionless plane. The coefficient of dynamic friction between the body and the plane will be

(a) $1/(1-n^2)$ (b) $1-(1/n^2)$

(c) $\sqrt{\{1-(1/n^2)\}}$ (d) $\sqrt{\{1/(1-n^2)\}}$

4. A small ball of mass m is allowed to slip down from the top of a hemispherical dome of radius R. At what height h from the lower end the ball will leave contact with the dome ?



(a) 2R / 3 (b) R / 2

(c) R / 4 (d) R / 6

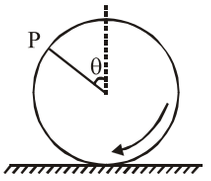
5. Three particles of equal mass m are situated at the vertices of an equilateral triangle of side L. What should be the velocity of each particle so that they move on a circular path without changing L?

(a) $\sqrt{(GM/2L)}$ (b) $\sqrt{(GM/L)}$

(c) $\sqrt{(2GM/L)}$ (d) $\sqrt{(GM/3L)}$

Space for Rough Work

6. A wheel is rolling straight on ground without slipping. If the axis of the wheel has speed v, the instantenous velocity of a point P on the rim, defined by angle θ, relative to the ground will be



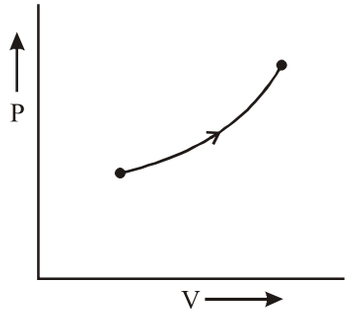
(a) $v\cos\left(\frac{1}{2}\theta\right)$ (b) $2v\cos\left(\frac{1}{2}\theta\right)$

(c) $v(1+\sin\theta)$ (d) $v(1+\cos\theta)$

7. A Carnot's engine works as a refrigerator between 250 K and 300 K. If it receives 750 calories of heat from the reservoir at the lower temperature, the amount of heat rejected at the higher temperature is

(a) 900 cal. (b) 625 cal. (c) 750 cal. (d) 1000 cal.

8. The figure shows a process on a gas in which pressure and volume both change. The molar heat capacity for this process is C. Then



(a) $C=0$ (b) $C=C_v$ (c) $C>C_v$ (d) $C<C_v$

9. A charge Q is distributed over two concentric hollow spheres of radii r and R (>r) such that the surface densities are equal and placed on the same axial points. Then the potential at the common centre is

(a) $\frac{Q(R^2+r^2)}{4\pi\varepsilon_0(R+r)}$ (b) $\frac{Q}{R+r}$

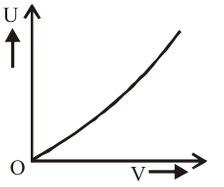
(c) Zero (d) $\frac{Q(R+r)}{4\pi\varepsilon_0(R^2+r^2)}$

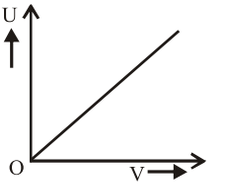
10. Suppose the gravitational force varies inversely as the nth power of the distance. Then the time period of a planet in circular orbit of radius R around the sun will be proportional to

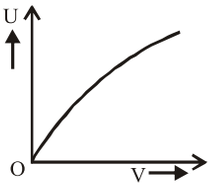
(a) R^n (b) R^{-n}

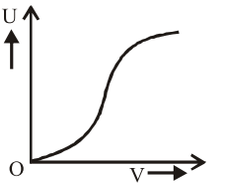
(c) $R^{(n+1)/2}$ (d) $R^{(n-1)/2}$

11. If a battery is connected across series combination of a capacitor and a resistor, at t = 0. If at an instant t potential difference across the capacitor be 'V' and energy stored in it be U. Then which of the following graph is correct.

(a)


(b)


(c)


(d)


12. Two different coils have self inductances L_1 = 8 mH and L_2 = 2 mH. The current in one coil is increased at a constant rate. The current in the second coils is also increased at the same constant rate. At a certain instant, the power given to the two coil is the same. At that instant of time, if W_1 and W_2 are the energies stored in the first and the second coil, respectively, the $W_1 : W_2$ is

(a) 1 : 2 (b) 2 : 1 (c) 4 : 1 (d) 1 : 4

13. A charge +q is taken around a long charged cylinder carrying a charge of λ per unit length. The total work done is

(a) zero (b) $4\pi\lambda/\varepsilon_0$ (c) λ/ε_0 (d) $\lambda/4\pi\varepsilon_0$

14. A cell of e.m.f. E is connected across a reistance R. The potential difference between the terminals of the cell is found to be V. The internal resistance of the cell must be

(a) [2(E–V)V]/R (b) [2(E–V)/V]R

(c) [(E–V)/V]R (d) (E–V)R

15. Two straight long conductors AOB and COD are perpendicular to each other and carry currents I_1 and I_2 respectively. The magnitude of the magnetic induction at a point P at a distance 'a' from the point O in a direction perpendicular to the plane ABCD is

(a) $\frac{\mu_0}{2\pi a}(I_1+I_2)$ (b) $\frac{\mu_0}{2\pi a}(I_1-I_2)$

(c) $\frac{\mu_0}{2\pi a}\left[I_1^2+I_2^2\right]^{1/2}$ (d) $\frac{\mu_0}{2\pi a}\left(\frac{I_1I_2}{I_1+I_2}\right)$

Space for Rough Work

16. The mutual inductance of a pair of coils, each of N turns, is M henry. If a current of I ampere in one of the coils is brought to zero in t second, the emf induced per turn in the other coil, in volt, will be

(a) $\frac{MI}{t}$ (b) $\frac{NMI}{t}$ (c) $\frac{MN}{It}$ (d) $\frac{MI}{Nt}$

17. A lens is placed between a source of light and a wall. It forms images of area A_1 and A_2 on the wall, for its two different positions. The area of the source of light is

(a) $\sqrt{(A_1A_2)}$ (b) $(A_1+A_2)/2$

(c) $\frac{\left(\sqrt{A_1}+\sqrt{A_2}\right)^2}{2}$ (d) $\left[\frac{1}{A_1}+\frac{1}{A_2}\right]^{-1}$

18. For sky wave propagation of a 10 MHz signal, what should be the minimum electron density in ionosphere

(a) $\sim 1.2\times10^{12}\,m^{-3}$ (b) $\sim 10^{6}\,m^{-3}$

(c) $\sim 10^{14}\,m^{-3}$ (d) $\sim 10^{22}\,m^{-3}$

19. In a npn transistor 10^{10} electrons enter the emitter in 10^{-6} s. 4% of the electrons are lost in the base. The current transfer ratio will be

(a) 0.98 (b) 0.97 (c) 0.96 (d) 0.94

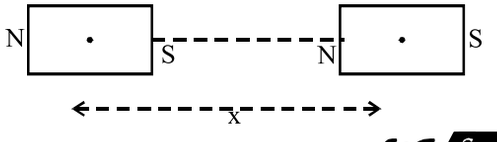
20. The vernier constant of two vernier callipers A and B are 0.01 cm and 0.01 mm respectively. Which one can measure the length of an object more accurately?

(a) Vernier A
(b) Vernier B
(c) Accuracy in measurement does not depend on vernier constant
(d) Both A and B are equally accurate.

PART-II (Numerical Answer Questions)

21. A mixture of two gases X and Y is enclosed, at constant temperature. The relative molecular mass of X, which is diatomic, is 8 times that of Y which is monoatomic. What is the ratio of the r.m.s speed of molecules of Y to that of molecules of X?

22. The mid points of two small magnetic dipoles of length d in end-on positions, are separated by a distance x, (x > > d). The force between them is proportional to x^{-n} where n is:



23. 56 tuning forks are so arranged in series that each fork gives 4 beats per sec with the previous one. The frequency of the last fork is 3 times that of the first. The frequency (in Hz) of the first fork is

24. Two slits separated by a distance of 1 mm are illuminated with red light of wavelength 6.5×10^{-7} m. The interference fringes are observed on a screen placed 1 m from the slits. The distance (in mm) between third dark fringe & the fifth bright fringe is equal to

25. The counting rate observed from a radioactive source at t = 0 second was 1600 counts per second and at t = 8 seconds it was 100 counts per second. The counting rate observed, as counts per second at t = 6 seconds will be

CHEMISTRY

PART-I (Multiple Choice Questions)

26. Which one of the following is an example of co-ordination isomerism ?

(a) $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$

(b) $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$

(c) $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2H_2O$

(d) $[Co(NH_3)_6][Cr(CN)_6]$ and $(Cr(NH_3)_6][Co(CN)_6]$

27. Two containers P and Q of equal volume (1 litre each) contain 6 g of O_2 and SO_2 respectively at 300 K and 1 atmosphere. then

(a) Number of molecules in P is less than that in Q.
(b) Number of molecules in P and Q is same.
(c) Number of molecules in Q is less than that in P.
(d) Either (a) or (b)

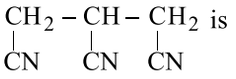
28. A wedding ring presented to a bride contains 788 mg of gold and rest diamond. If the ring weighs 1 g the bride receives (At wt : Au = 197, C = 12)

(a) more number of gold atoms
(b) more number of carbon atoms
(c) equal number of gold and carbon atoms
(d) gold and carbon atoms in ratio of 4:1 approximately

29. Urea and hydrazine react to form ammonia gas along with compound X which reacts with aldehydes and ketones to form specific crystalline derivatives. X is

(a) phenylhydrazine (b) semicarbazide

(c) biuret (d) acetylurea

30. The I.U.P.A.C. name of  is

(a) 1, 2, 3-propanetricarbonitrile

(b) 1, 2, 3 – tricyanopropane

(c) 3-cyano-1, 5-dinitrilepentane

(d) 3-cyano – 1, 5-pentanedinitrile

31. Diborane has

(a) 2 two - centre - two - electron bonds and 4 three - centre - three - electron bonds

(b) 4 two - centre - two - electron bonds and 2 three - centre - three - electron bonds

(c) 3 two - centre - two - electron bonds and 3 three - centre - three - electron bonds

(d) all six identical bonds.

32. A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour. Y is

(a) $Mg(NO_3)_2$ (b) Mg_3N_2

(c) NH_3 (d) MgO

33. Which of the following has a shape different from others?

(a) $[Zn(NH_3)_4]^{+2}$ (b) $Ni(CO)_4$

(c) $[Cd(CN)_4]^{-2}$ (d) $[Cu(NH_3)_4]^{+2}$

34. Di-n-propyl ether and diallyl ether can be distinguished by

(a) Acetic acid

(b) Sodium Metal

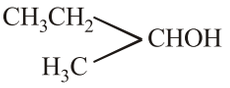
(c) Cold dilute $KMnO_4$ solution

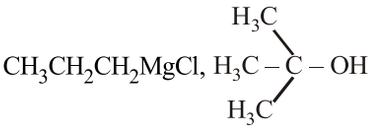
(d) PCl_5

35. A and B in the reaction,

$$CH_3CH_2CH_2Cl \xrightarrow[\text{ether}]{Mg} A \xrightarrow[2.H_3O^+]{1.HCHO} B$$

(a) $CH_3CH_2CH_2MgCl$, $CH_3CH_2CH_2CH_2OH$

(b) $CH_3CHMgCH_2Cl$, 

(c) $CH_3CH_2CH_2MgCl$, 

(d) $CH_3CH_2CH_2MgCl$, $CH_3CH_2OCH_2CH_3$

36. Alcohols can be obtained from all methods except

(a) hydroboration-oxidation

(b) oxymercuration-demercuration

(c) reduction of aldehyde/ketones with Zn–Hg/HCl

(d) fermentation of starch

37. Which of the following expressions is not true ?

(a) For a netural solution $[H^+] = [OH^-] = \sqrt{K_w}$

(b) For an acidic solution

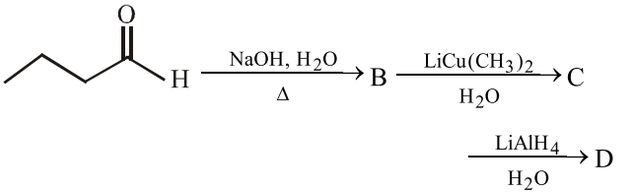
$[H^+] > \sqrt{K_w}$ and $[OH^-] < \sqrt{K_w}$

(c) For an alkaline solution

$[H^+] < \sqrt{K_w}$ and $[OH^-] > \sqrt{K_w}$

(d) For a neutral solution $[H^+] = [OH^-] = 10^{-7}$ M at all temperatures

38. The final product of the following sequence of reactions



(a) 3-ethyl-2-methyl-1-hexanol

(b) 2, 3-dimethyl-1-pentanol

(c) 2-ethyl-3-methyl-1-hexanol

(d) 3, 3-dimethyl-1-pentanol

39. A solution containing As^{3+}, Cd^{2+}, Ni^{2+} and Zn^{2+} is made alkaline with dilute NH_4OH and treated with H_2S. The precipitate obtained will consist of

(a) As_2S_3 and CdS (b) CdS, NiS and ZnS

(c) NiS and ZnS (d) Sulphide of all ions

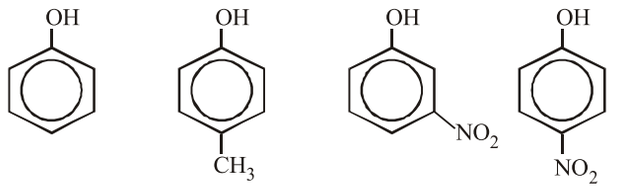
40. M_2SO_4 (M^+ is a monovalent metal ion) has a K_{sp} of 3.2×10^{-5} at 298 K. The maximum concentration of SO_4^{-2} ion that could be attained in a saturated solution of this solid at 298 K is

(a) 3×10^{-3} M (b) 7×10^{-2} M

(c) 2.89×10^{-4} M (d) 2×10^{-2} M

Space for Rough Work

41. In the following compounds,



(I) (II) (III) (IV)

the order of acidity is

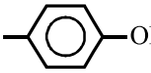
(a) III > IV > I > II (b) I > IV > III > II

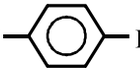
(c) II > I > III > IV (d) IV > III > I < II

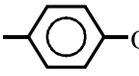
42. The electrolyte most effective for bringing about coagulation of Fe $(OH)_3$ sol is

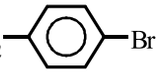
(a) $K_4[Fe(CN)_6]$ (b) $BaCl_2$

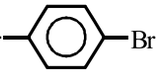
(c) NaCl (d) None

43. The reaction of CH_3 CH = CH  OH with HBr gives

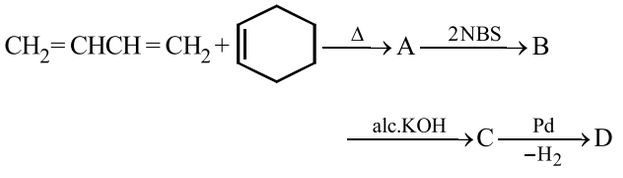
(a) $CH_3CH=CH$  Br.

(b) CH_3CH_2CHBr  OH

(c) $CH_3CHBrCH_2$  Br

(d) CH_3CH_2CHBr  Br

44. Final product (D) formed in the following reaction sequence is



(a) Naphthalene (b) Tetralin

(c) Benzene (d) Anthracene

45. A current of 2.0A when passed for 5 hours through a molten metal salt deposits 22.2g of metal of atomic weight 177. The oxidation state of the metal in the metal salt is

(a) +1 (b) +2 (c) +3 (d) +4

PART-II (Numerical Answer Questions)

46. A gaseous mixture of three gases A, B and C has a pressure of 10 atm. The total number of moles of all the gases is 10. If the partial pressures of A and B are 3.0 and 1.0 atm respectively and if 'C' has mol. wt of 2.0 what is the weight of 'C' in g present in the mixture.

47. For a chemical reaction $X \rightarrow Y$, the rate of reaction increases by a factor of 1.837 when the concentration of X is increased by 1.5 times. What is the order of the reaction with respect to X ?

48. The standard reduction potential for Cu^{2+}/Cu is + 0.34. Calculate the reduction potential at pH = 14 for the above couple. (K_{sp} $Cu(OH)_2 = 1 \times 10^{-19}$)

49. The solution containing 4.0 % PVC in one litre of dioxane was found to have osmotic pressure of 6.0×10^{-2} atm at 300 K. What is the molecular mass of polymer ?

50. The γ-form of iron has *fcc* structure (edge length 386 pm) and β-form has *bcc* structure (edge length 290 pm). The ratio of density of γ-form and β-from is

MATHEMATICS

PART-I (Multiple Choice Questions)

51. ${}^{n+4}C_r - {}^nC_r - 3.{}^nC_{r-1} - 3.{}^nC_{r-2} - {}^nC_{r-3}$ is equal to

(a) ${}^{n+1}C_{r-1}$ (b) ${}^{n+2}C_{r-1}$

(c) ${}^{n+3}C_{r-1}$ (d) ${}^{n+4}C_{r-1}$

52. The sum of the first three consecutive terms of an A.P. is 9 and the sum of their squares is 35 . The sum to n terms of the series is

(a) $3n^2$ (b) $2n^2$

(c) $6n - 2n^2$ (d) $6n - n^2$

53. Let $\alpha/(\alpha-1)$ and $\beta/(\beta-1)$ be the roots of $x^2 + ax + b = 0$. Then $1/\alpha$ and $1/\beta$ are the roots of

(a) $bx^2 + ax + 1 = 0$

(b) $bx^2 - ax + 1 = 0$

(c) $bx^2 + (a + 2b)x + a + b + 1 = 0$

(d) $bx^2 - (a + 2b)x + a + b + 1 = 0$

54. If tan(A/2), tan(B/2), tan(C/2) are in A.P. then sec A, sec B, sec C are in

(a) A.P. (b) G.P. (c) H.P. (d) None

55. The straight line y = mx + c is a tangent to the parabola $x^2 = 4by$ if

(a) $c = b/m$ (b) $c = -bm$ (c) $c = -bm^2$ (d) $c = \frac{b}{m^2}$

56. If tangents be drawn to the cricle $x^2 + y^2 = 12$ at its points of intersection with the circle $x^2 + y^2 - 5x + 3y - 2 = 0$, then the tangents intersect at the point

(a) $\left(-6, \frac{18}{5}\right)$ (b) $\left(6, \frac{18}{5}\right)$

(c) $\left(-6, -\frac{18}{5}\right)$ (d) $\left(6, -\frac{18}{5}\right)$

57. The locus of the mid points of the chords of the ellipse $x^2/a^2 + y^2/b^2 = k$, $k > 0$, making equal intercepts on the coordinate axes, is

(a) $x = y$ (b) $x + y = 0$

(c) $x/a^2 = y/b^2$ (d) $x/a^2 + y/b^2 = 0$

58. I= n/m means that n is a factor of m, then the relation 'I', is

(a) Reflexive and symmetric

(b) Transitive and symmetric

(c) Reflexive, transitive and symmetric

(d) Reflexive, transitive and not symmetric

59. The value of $\lim_{x \to \pi/6} (4 - 3\sin x - 2\cos^2 x)^{\frac{1}{2\sin x - 1}}$ is

(a) 1 (b) e (c) $\sqrt{e}$ (d) $e^{-1/2}$

60. Let $f(x) = (x-1)^2 (x-2)^3 e^x$. Then

(a) f(x) has local maximum at x = 1

(b) f(x) has point of inflexion at x = 1

(c) f(x) has local minimum at x = 2

(d) all of these

61. If $2a + 3b + 6c = 0$ then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval

(a) (0, 1) (b) (1, 3) (c) (2, 3) (d) (1, 2)

62. A curve y = f(x) passes through the point P(1, 1). The normal to the curve at P is a (y – 1) + (x – 1) = 0. If the slope of the tangent at any point on the curve is proportional to the ordinate of the point, then the equation of the curve is

(a) $y = e^{a(x-1)}$ (b) $y = e^{a(1-x)}$

(c) $y = e^{a/2(x-1)}$ (d) $e^{a/2(x+1)}$

63. Value of $\int_0^{\pi} \sin^n x \cos^{2m+1} x \, dx$

(where m, n ∈ N) is

(a) $\frac{(2m+1)!}{n!}$ (b) $\frac{(2m+1)!}{n^2}$

(c) $\int_0^{\pi} \cos^{2m+1} x \, dx$ (d) none of these

64. The primitive of $\frac{\sin x}{\sin 4x}$ is

(a) $\frac{1}{4}\log\left|\frac{\sin x - 1}{\sin x + 1}\right| - \frac{1}{\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x - 1}{\sqrt{2}\sin x + 1}\right| + C$

(b) $\frac{1}{8}\log\left|\frac{\cos x - 1}{\cos x + 1}\right| - \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}\cos x - 1}{\sqrt{2}\cos x + 1}\right| + C$

(c) $\frac{1}{8}\log\left|\frac{\sin x - 1}{\sin x + 1}\right| - \frac{1}{4\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x - 1}{\sqrt{2}\sin x + 1}\right| + C$

(d) none of these

65. Let coordinates of a point 'p' with respect to the system of non-coplanar vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is (3, 2, 1). Then coordinates of 'p' with respect to the system of vectors $\vec{a} + \vec{b} + \vec{c}$, $\vec{a} - \vec{b} + \vec{c}$ and $\vec{a} + \vec{b} - \vec{c}$ is

(a) $\left(\frac{3}{2}, \frac{1}{2}, 1\right)$ (b) $\left(\frac{3}{2}, 1, \frac{1}{2}\right)$

(c) $\left(\frac{1}{2}, \frac{3}{2}, 1\right)$ (d) None

66. The three lines through the origin with direction cosines l_1, m_1, n_1; l_2, m_2, n_2; and l_3, m_3, n_3 are coplanar, if

(a) $l_1(l_2 - l_3) + m_1(m_2 - m_3) + n_1(n_2 - n_3) = 0$

(b) $l_1(m_2 n_3 - m_3 n_2) + m_1(n_2 l_3 - n_3 l_2) + n_1(l_2 m_3 - l_3 m_2) = 0$

(c) $\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$

(d) None of these

Space for Rough Work

67. If $x + 2 = 6$, then $x = 4$. So, which statement is its converse?

(a) If $x \neq 4$, then $x + 2 \neq 6$

(b) If $x = 4$, then $x + 2 \neq 6$

(c) If $x = 4$, then $x + 2 = 6$

(d) If, $x \neq 4$ then $x + 2 = 6$

68. If $y = f(x)$ be a monotonically increasing or decreasing function of x and M is the median of variable x, then the median of y is

(a) f(M) (b) M/2 (c) $f^{-1}(M)$ (d) None

69. If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and λ is a real number then $[\lambda(\vec{a} + \vec{b})\ \lambda^2\vec{b}\ \lambda\vec{c}] = [\vec{a}\quad \vec{b} + \vec{c}\quad \vec{b}]$ for

(a) exactly one value of λ

(b) no value of λ

(c) exactly three values of λ

(d) exactly two values of λ

70. A and B are two independent events. The probability that both A and B occur is 1/6 and the probability that neither of them occurs is 1/3. The probability of occurrence of A is :

(a) $\frac{5}{6}$ (b) $\frac{1}{6}$

(c) $\frac{1}{5}$ (d) None of these

PART-II (Numerical Answer Questions)

71. For any real θ, if the maximum value of $\cos^2(\cos\theta) + \sin^2(\sin\theta)$ is $a + \sin^2 b$, the $a + b =$

72. Let $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$ and $(10)B = \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$

If B is the inverse of matrix A, then α is

73. The derivative of $\tan^{-1}[(3x^2 - 1)/(3x - x^3)]$ with respect to $\sin^{-1}[(x^2 - 1)/(x^2 + 1)]$ is

74. If $f'(x) = f(x)$ and $f(0) = 2$, if $\int \frac{f(x)}{3 + 4f(x)}dx$

$= a\log(b + ce^x) +$ constant, then $a(b + c) =$

75. If X and Y are independent binomial variates $B\left(5, \frac{1}{2}\right)$ and $B\left(7, \frac{1}{2}\right)$, then $\frac{1}{p(x + y = 3)}$

RESPONSE SHEET

PHYSICS

1.	ⓐⓑⓒⓓ	2.	ⓐⓑⓒⓓ	3.	ⓐⓑⓒⓓ	4.	ⓐⓑⓒⓓ	5.	ⓐⓑⓒⓓ
6.	ⓐⓑⓒⓓ	7.	ⓐⓑⓒⓓ	8.	ⓐⓑⓒⓓ	9.	ⓐⓑⓒⓓ	10.	ⓐⓑⓒⓓ
11.	ⓐⓑⓒⓓ	12.	ⓐⓑⓒⓓ	13.	ⓐⓑⓒⓓ	14.	ⓐⓑⓒⓓ	15.	ⓐⓑⓒⓓ
16.	ⓐⓑⓒⓓ	17.	ⓐⓑⓒⓓ	18.	ⓐⓑⓒⓓ	19.	ⓐⓑⓒⓓ	20.	ⓐⓑⓒⓓ
21.		22.		23.		24.		25.	

CHEMISTRY

26.	ⓐⓑⓒⓓ	27.	ⓐⓑⓒⓓ	28.	ⓐⓑⓒⓓ	29.	ⓐⓑⓒⓓ	30.	ⓐⓑⓒⓓ
31.	ⓐⓑⓒⓓ	32.	ⓐⓑⓒⓓ	33.	ⓐⓑⓒⓓ	34.	ⓐⓑⓒⓓ	35.	ⓐⓑⓒⓓ
36.	ⓐⓑⓒⓓ	37.	ⓐⓑⓒⓓ	38.	ⓐⓑⓒⓓ	39.	ⓐⓑⓒⓓ	40.	ⓐⓑⓒⓓ
41.	ⓐⓑⓒⓓ	42.	ⓐⓑⓒⓓ	43.	ⓐⓑⓒⓓ	44.	ⓐⓑⓒⓓ	45.	ⓐⓑⓒⓓ
46.		47.		48.		49.		50.	

MATHEMATICS

51.	ⓐⓑⓒⓓ	52.	ⓐⓑⓒⓓ	53.	ⓐⓑⓒⓓ	54.	ⓐⓑⓒⓓ	55.	ⓐⓑⓒⓓ
56.	ⓐⓑⓒⓓ	57.	ⓐⓑⓒⓓ	58.	ⓐⓑⓒⓓ	59.	ⓐⓑⓒⓓ	60.	ⓐⓑⓒⓓ
61.	ⓐⓑⓒⓓ	62.	ⓐⓑⓒⓓ	63.	ⓐⓑⓒⓓ	64.	ⓐⓑⓒⓓ	65.	ⓐⓑⓒⓓ
66.	ⓐⓑⓒⓓ	67.	ⓐⓑⓒⓓ	68.	ⓐⓑⓒⓓ	69.	ⓐⓑⓒⓓ	70.	ⓐⓑⓒⓓ
71.		72.		73.		74.		75.	